Embedded pile rows in Tellus 2D: element verification, calibration against a piled abutment case under axial and lateral loading, and an updated mesh option

Tellus 2D development report

12 November 2026

Abstract

This report checks the embedded pile row of Tellus 2D under axial and lateral loading and documents the updated mesh option that was added for the purpose. On the element level, a pile row in a rigid soil reproduces the closed-form axial and lateral stiffness and the axial and lateral capacities within 1 %. As a calibration case I reran the PLAXIS case study of a piled bridge abutment in an embankment on soft soil, in which three raked pile rows are loaded axially by the abutment and by negative skin friction and laterally by the soil moving towards the cut under the deck. With small displacements and 150 days of consolidation after construction, Tellus gives pile forces of the same sign, shape and position as the PLAXIS 2D and 3D results, with a largest normal force of −905 kN per pile in row 2 against −1188 kN (PLAXIS 2D) and −1053 kN (PLAXIS 3D), and largest moments of 75 and 89 % of the PLAXIS 3D values in rows 1 and 2 and about half in row 3. The comparison is not yet like for like, because the source consolidates until the excess pore pressure is below 5 kPa with an updated mesh, whereas the Tellus run still has 93 kPa of excess pore pressure and its pile forces are still rising. The updated mesh option is verified on a large-strain column and on the elastica, but its run of the abutment case stalled when the piles were installed, which is an open defect.

1 Purpose and scope

Tellus 2D has had embedded pile rows since October 2026, checked so far only for their axial capacity. Piles in practice also carry lateral load, from a head force or from soil that moves past them, so the element needs a check in both directions before it is used in design. This report gives two levels of evidence. The first is a verification of the element itself against closed-form solutions, which shows whether the springs and the beam are coded correctly. The second is a calibration against a published case in which pile rows carry axial and lateral load together, which shows whether the whole model, with soft soil, consolidation and staged construction, gives the pile forces that the reference program gives. No field measurement is involved in either level, so the report says nothing about how well any of the programs predicts a real pile.

2 The embedded pile row in Tellus 2D

A pile row is a line of beam elements with their own nodes, which lies over the soil mesh without splitting it, so the soil can move between the piles. Each pile node is joined to the soil node under it by an axial spring, a lateral spring and, at the tip, a base spring, and all properties of one pile are divided by the out-of-plane spacing L_s to give values per metre run. The spring

stiffnesses follow Sluis (2012) and the PLAXIS 2D implementation,

$$R_S = ISF_{RS}\,\frac{G}{L_s}, \qquad R_N = ISF_{RN}\,\frac{G}{L_s}, \qquad K_F = ISF_{KF}\,\frac{G\,R_{eq}}{L_s}, \tag{1}$$

where G is the shear modulus of the soil at the node and R_{eq} the equivalent pile radius, with the default factors

$$ISF_{RS} = ISF_{RN} = 2.5\left(\frac{L_s}{D_{eq}}\right)^{-0.75}, \qquad ISF_{KF} = 25\left(\frac{L_s}{D_{eq}}\right)^{-0.75}. \tag{2}$$

The axial spring slides when it reaches the skin resistance T_{max}/L_s, the base spring acts in compression only up to F_{max}/L_s, and the lateral spring is elastic unless a lateral limit is given. The defaults were derived for bored piles, static loading and L_s/D_{eq} between 2 and 8, so outside that range the factors need their own calibration.

3 Element verification against closed-form solutions

The first check isolates the element from the soil. One vertical pile row ($D = 0.6$ m, $L = 12$ m, $E = 30$ GPa, $L_s = 2$ m) stands in a soil that is made almost rigid ($G = 10^6$ MPa), and the interface factors are chosen so that the springs have round values per metre run, 50 MN/m^2 along the shaft, 20 MN/m^2 across it and 150 MN/m at the base. The pile then is a beam on Winkler springs, for which the axial response is the hyperbolic solution of a bar with shaft and base springs and the lateral response is the Hetényi solution of a finite beam loaded at its free head. The capacities follow from statics, which is the integral of the skin resistance plus the base capacity for axial loading, and $H_u = (\sqrt{2}-1)\,p_u L$ for a rigid free-head pile with a uniform lateral limit p_u.

Table 1 shows that Tellus reproduces every closed-form value within 1 %. The axial stiffness agrees to 0.02 %, whereas the lateral deflection is 0.6 % larger and the moment 0.6 % smaller than the Hetényi values, which I attribute to the shear flexibility of the Mindlin beam because the closed form is for a beam without shear deformation. The capacities are 0.5 to 0.8 % above the exact values, which is the size of the load step at which the last converged state is found. The script is `verify_element.py`.

Table 1: Element verification of the embedded pile row in a rigid soil (values per metre run).

Check	Quantity	Tellus 2D	Closed form	Difference [%]
Axial stiffness	head settlement [mm]	0.2341	0.2341	+0.02
	base force [kN/m]	13.87	13.87	+0.00
	skin traction at the head [kN/m^2]	11.7	11.71	-0.00
Lateral stiffness	head deflection [mm]	2.407	2.392	+0.62
	largest moment [kNm/m]	33.48	33.69	-0.63
Axial capacity	compression [kN/m]	504.1	500	+0.83
	tension [kN/m]	302.3	300	+0.78
Lateral capacity	free head, uniform limit [kN/m]	74.92	74.56	+0.49

4 Calibration case: a piled bridge abutment in an embankment on soft soil

4.1 Source and reference data

The case is the PLAXIS case study *Embedded Pile Row in Plaxis 2D* (November 2012), written when the feature was released and based on the MSc thesis of Sluis (2012). It models a bridge abutment on three pile rows inside an approach embankment on about 12 m of clay, peat and clayey sand over deep sand, which is a common Dutch situation. The piles are loaded axially by the abutment and by the soft layers that keep settling around them, and laterally by the soil that moves towards the cut under the bridge deck, so one model exercises the skin, base and lateral springs together. The source gives the normal force, shear force and bending moment along each pile from PLAXIS 2D with embedded pile rows and from PLAXIS 3D with embedded piles at their real positions (its Figures A.7 to A.9). These charts are vector graphics in the PDF, so I read the marker positions from the drawing commands, which gives the curves to the plotting accuracy of the source, 180 points per 2D curve and about 300 per 3D curve. The values are per pile, which for the 2D results means the plane-strain force times the 2.4 m spacing. I could not obtain the thesis itself, because the TU Delft repository did not answer during this work, so the simpler single-pile-row tests of the thesis are not part of this report.

4.2 Model

Figure 1 shows the model. The source gives the layer levels, the soil and structural parameters and the pile coordinates in its text and tables, and it gives the outline of the embankment, the excavation and the abutment only as figures, so I scaled these from its Figures 5 and 6 using the pile heads as the length reference. The model is 50 m wide and reaches from +7.2 to −22 m NAP, with the water level at −0.5 m. The sands use Hardening Soil ($E_{50}^{ref} = E_{oed}^{ref} = 30$ MPa, $E_{ur}^{ref} = 90$ MPa, $m = 0.5$, $c' = 1$ kPa, $\varphi' = 30°$), and the clay, peat, sandy clay and clayey sand use Soft Soil Creep with undrained (A) behaviour and the parameters of Table 3 of the source. The abutment is an elastic plate ($EA = 2 \times 10^7$ kN/m, $EI = 1.67 \times 10^6$ kNm^2/m, $w = 25$ kN/m/m), and the three pile rows are 0.54 m solid circular piles at 2.4 m centres with $E = 20$ GPa, a skin resistance rising linearly from 10 to 100 kN/m, a base capacity of 1000 kN, the default interface stiffness factors and a rigid connection to the slab. Rows 1 and 2 rake forwards and row 3 rakes backwards.

The construction follows Table 7 of the source, with every stage a consolidation phase in which the change is applied over the phase time. The embankment rises in three lifts of 2 days with 30, 60 and 90 days of consolidation, the abutment pit is excavated in 2 days, the abutment and the piles are installed over 10 days, and in 5 days the sand under the deck is removed and the fill behind the wall is placed. The mesh has 1,833 15-node triangles and 15,497 nodes. Table 2 lists the phases as calculated.

4.3 Differences from the source model

Four differences from the source remain, and the first two matter for the comparison. First, the source uses an updated mesh in every phase, whereas the Tellus result reported here is a small-displacement run, because the updated mesh run stalled (Section 5.3). Second, the source ends with a consolidation phase that stops when the excess pore pressure is below 5 kPa, whereas the Tellus solver takes at most 250 time steps per phase and needed about one day per step in the last phases, so the run reported here ends 150 days after construction (day 353) with a largest excess pore pressure of 93 kPa under the full embankment and 7 kPa of suction near the cut.

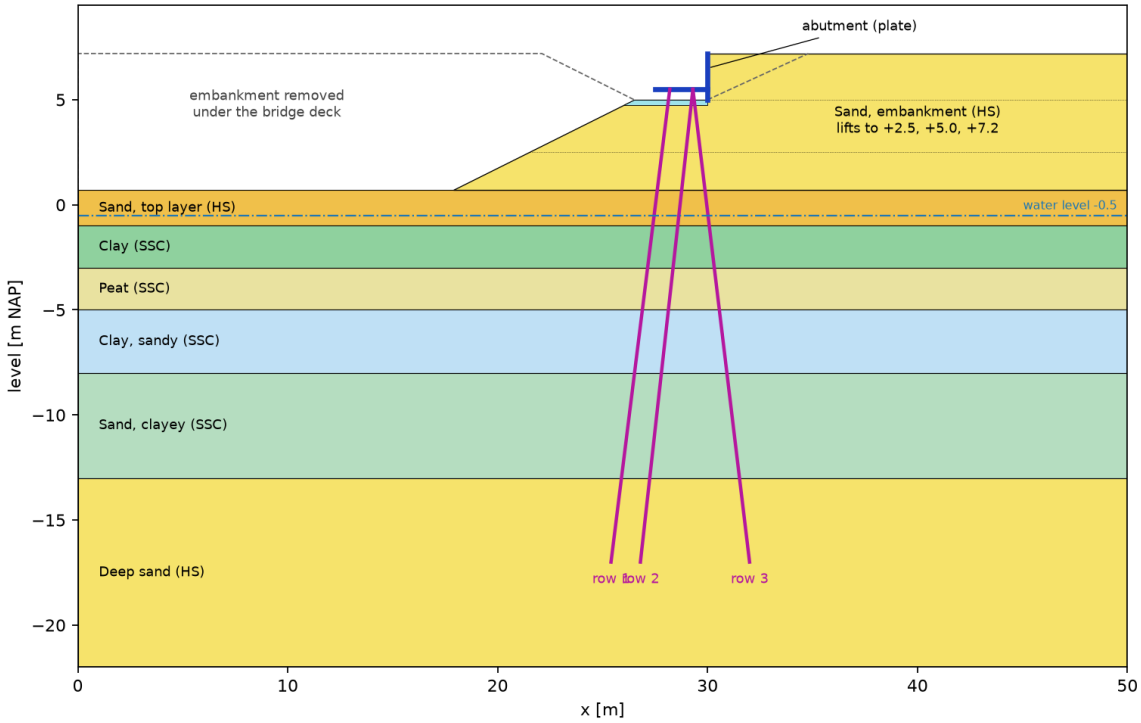


Figure 1: The piled abutment model in Tellus 2D at its final stage. The dashed outline is the embankment that is built over the full width and later removed under the bridge deck.

Third, the 0.25 m layer of sand with $c' = 5$ kPa under the abutment has that material from its placement, whereas the source swaps it in when the abutment is installed. Fourth, the interface along the wall is not extended below the wall toe. The geometry scaled from the figures adds an uncertainty of about 0.1 m in the positions of the slopes.

4.4 Results

Figure 2 and Table 3 compare the pile forces at day 353 with the two PLAXIS results. The Tellus curves have the shape of the reference curves in all nine panels. The bending moments change sign at the same levels, near −5 m NAP where the piles pass from the soft layers into the stiffer clayey sand, the largest positive moments sit at −8 to −9 m in all three programs, and the normal force in rows 1 and 2 peaks between −6 and −9 m, in the lower part of the soft layers, which is where negative skin friction turns into positive friction. Row 3, which rakes backwards, is in tension at its head in Tellus (309 kN) as in PLAXIS 2D (366 kN) and PLAXIS 3D (238 kN at the head, 526 kN at a local peak).

The magnitudes are lower than the reference in most panels. In row 2, the most heavily loaded row, the largest normal force is −905 kN per pile against −1188 kN in PLAXIS 2D and −1053 kN in PLAXIS 3D, so Tellus is at 76 % and 86 % of the two references, and the root-mean-square difference along the pile is 19 % and 7 % of the reference peaks. In row 1 the largest normal force is −331 kN against −336 and −443 kN. The moments in rows 1 and 2 below +2 m follow both PLAXIS curves closely, with a root-mean-square difference of 15 to 20 %, whereas in row 3 the Tellus moments and shear forces are about half of the reference values (largest moment 290 kNm against 640 and 593 kNm). Near the pile heads Tellus gives a much smaller moment than PLAXIS in all three rows, about 70 kNm against 324 kNm (PLAXIS 2D) and 253 kNm (PLAXIS 3D) in row 1.

Table 2: Phases of the abutment case in Tellus 2D (small displacements).

No.	Phase	Converged	Solver message
0	Initial phase	yes	K0 procedure
1	emb. layer 1	yes	11 time steps to t = 2.0000 day, 7620 plastic points
2	cons. layer 1	yes	16 time steps to t = 32.0000 day, 6507 plastic points
3	emb. layer 2	yes	11 time steps to t = 34.0000 day, 10920 plastic points
4	cons. layer 2	yes	18 time steps to t = 94.0000 day, 10138 plastic points
5	emb. layer 3	yes	9 time steps to t = 96.0000 day, 14074 plastic points
6	cons. layer 3	yes	19 time steps to t = 186.0000 day, 13296 plastic points
7	excavate for abutment	yes	10 time steps to t = 188.0000 day, 5157 plastic points
8	install abutment and piles	yes	13 time steps to t = 198.0000 day, 6850 plastic points
9	remove sand beneath deck, fill behind abutment	yes	11 time steps to t = 203.0000 day, 7039 plastic points
10	consolidate 50 d	yes	49 time steps to t = 253.0000 day, 4668 plastic points
11	consolidate 150 d	yes	182 time steps to t = 353.0000 day, 4745 plastic points
12	consolidate 300 d	no	maximum number of steps reached
13	consolidate 600 d	no	not calculated: 'consolidate 300 d', which it starts from, did not fin
14	consolidate 1,200 d	no	not calculated: 'consolidate 600 d', which it starts from, did not fin

Two causes explain the direction of these differences, and the present run cannot separate them. The main one is time, because the pile forces in Tellus rise from one result phase to the next, in row 2 from -719 kN at the end of construction to -845 kN after 50 days and -905 kN after 150 days, and the soil at the abutment is still moving, whereas the reference is the state after full consolidation. The second is the updated mesh of the source, which adds second-order moments in piles that have moved sideways. The small head moments are a separate point that needs its own check, because they suggest that the rigid connection between the pile heads and the slab, or the 0.5 m of pile that stands free between the slab and the soil, behaves differently in the two programs. With these reservations, the result supports the element and its default interface stiffness factors in combined axial and lateral loading, because a wrong lateral or axial spring would shift the levels of the peaks and of the sign changes, and these agree. It does not yet establish the magnitudes.

5 Updated mesh for large displacements

5.1 Formulation

The source calculates every phase with an updated mesh, which Tellus 2D did not have, so the option was added as part of this work. A plastic or consolidation phase with the option writes equilibrium on the deformed mesh in an updated Lagrangian form. At each stress point the shape function gradients on the deformed mesh follow from those on the mesh as generated through the deformation gradient, $\partial N/\partial x = F^{-T}\,\partial N/\partial X$ with $F = I + \sum_a u_a \otimes \partial N_a/\partial X$. The strain and rotation increments of a step use the gradients at the middle of the step, the

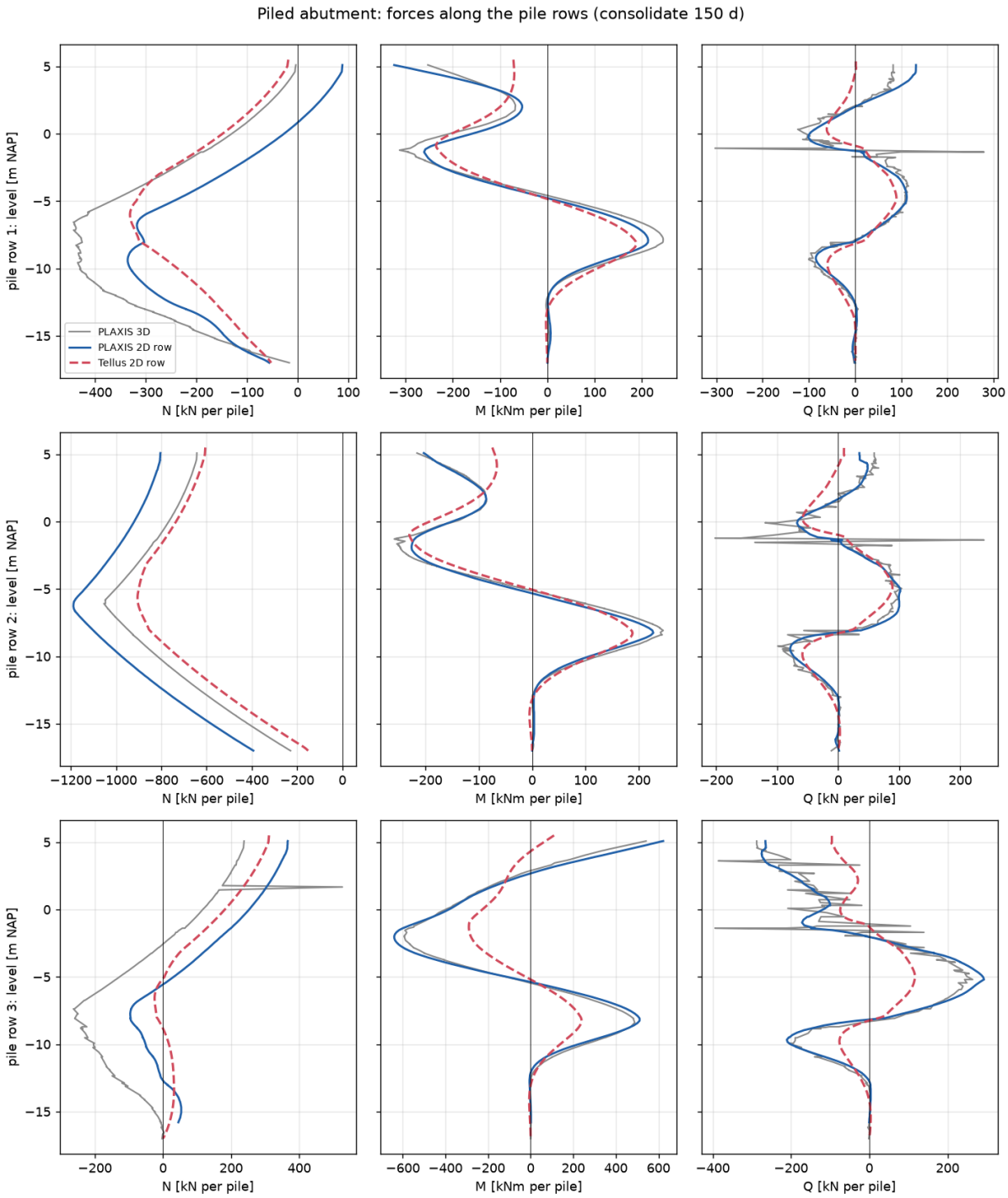


Figure 2: Normal force, bending moment and shear force along the three pile rows, per pile (the 2D values times the 2.4 m spacing): Tellus 2D against the PLAXIS 2D embedded pile row and PLAXIS 3D results of the source. Tellus result phase: consolidate 150 d.

Table 3: Largest pile forces per pile with their level in m NAP in brackets, and the root-mean-square difference along the pile between Tellus and each PLAXIS curve as a share of the largest PLAXIS value.

Row	Force	Tellus 2D	PLAXIS 2D	PLAXIS 3D	rms vs 2D [%]	rms vs 3D [%]
1	N [kN]	-331 (-6.0)	-336 (-9.3)	-443 (-6.6)	28	22
	M [kNm]	-236 (-0.9)	-307 (+5.0)	-313 (-1.2)	16	15
	Q [kN]	90 (-4.7)	132 (+5.0)	-303 (-1.1)	32	13
2	N [kN]	-905 (-5.6)	-1188 (-6.3)	-1053 (-6.1)	19	7
	M [kNm]	-230 (-1.0)	227 (-8.3)	-258 (-1.3)	20	18
	Q [kN]	89 (-4.9)	102 (-5.0)	239 (-1.4)	25	16
3	N [kN]	309 (+5.0)	366 (+5.0)	526 (+1.7)	16	26
	M [kNm]	-290 (-1.2)	-640 (-2.0)	-593 (-1.9)	36	33
	Q [kN]	117 (-5.0)	293 (-5.2)	-394 (-1.4)	40	26

stress at the start of the step is turned by the rotation increment (Hughes and Winget, 1980), and the internal forces use the gradients at the end of the step with the Kirchhoff stress, so the integration volume stays that of the original mesh. The stiffness matrix is rebuilt on the deformed mesh when the displacement gradient of any element has changed by more than 0.03 since the last factorisation.

Plates and embedded piles become co-rotational beams, in which each element deforms in a frame that turns with its chord and its forces turn back with that frame, so an axial force acts with the lever arm of the deflection. A geogrid takes its strain from the stretch of its deformed line and pulls along its deformed tangent, which gives membrane action. A phase that starts from an updated mesh phase is an updated mesh phase too. The soil weight stays that of the mesh as generated, the pore pressures stay those of the points as generated, and interfaces, the springs of embedded piles and anchors keep their original directions, which are the present limits of the option.

5.2 Verification

Table 4 gives the soil check, a weightless linear elastic column 2 m high under a uniform pressure q with its sides on rollers. Small-displacement theory gives a settlement qH/E_{oed}, whereas with the Kirchhoff stress and the logarithmic strain the stretch λ solves $E_{oed} \ln \lambda = -q\lambda$. Tellus follows this solution within 0.5 % at 16 % and at 30 % compression in four load steps, and the option changes nothing at small load. The beam check is the elastica, where a cantilever with $PL^2/EI = 1$ deflects $0.3017\,L$ and shortens by $0.0564\,L$ (Mattiasson, 1981), and the co-rotational element reproduces both within the test tolerances of $0.002\,L$ and $0.001\,L$. A beam turned as a rigid body by 30° carries no force, and small models of a pile row and a plate in elastic soil give the same answer with and without the option within 0.5 %.

Table 4: Updated mesh: settlement of a 2 m elastic column under a uniform pressure against the closed-form values.

Run	Tellus 2D [m]	Reference [m]	Difference [%]
small load	0.0020	0.0020	-0.00
small load, updated mesh	0.0020	0.0020	-0.00
q = 0.2 E_{oed}	0.4000	0.4000	-0.00
q = 0.2 E_{oed}, updated mesh	0.3123	0.3108	+0.46
q = 0.5 E_{oed}, updated mesh	0.5906	0.5931	-0.41

5.3 Status on the abutment case

The updated mesh run of the abutment case passed the three embankment lifts, their consolidation and the excavation, seven phases with settlements of about 0.4 m, at a pace similar to the small-displacement run. It then stalled in the first time step of the phase that installs the abutment and the piles, with no convergence message, and I stopped it. Small models with a pile row, a plate, or both rigidly joined run without difficulty, so the defect needs structures that are placed in a mesh that has already deformed, and I have not found its cause. Until it is fixed, the updated mesh should be treated as verified for soil and for structures placed in an undeformed mesh, and as not usable for a case like this one.

6 Conclusions and open items

The embedded pile row of Tellus 2D is correct on the element level, where its axial and lateral stiffness and capacities agree with closed-form solutions within 1 %. In the piled abutment case it gives pile forces of the same sign, shape and position as PLAXIS 2D and PLAXIS 3D, and magnitudes of 76 to 99 % of the PLAXIS 2D normal forces and 45 to 100 % of its moments at a stage that is earlier than the reference state. I regard the element as verified and the case as a partial calibration, which is weaker than the agreement within a few percent that the same program reached on its other calibration cases.

Four items are open. The abutment case must be run to an excess pore pressure below 5 kPa, which needs a chain of consolidation phases or a higher step limit, and the slow convergence of about 35 iterations per step in those phases deserves a look of its own. The updated mesh defect at the installation of structures in a deformed mesh must be found and fixed, after which the case can be rerun as the source ran it. The pile head moments need a check on a small model with a rigid connection and a free length of pile. The solver regression suite was interrupted during this work and must be run in full on the changed solver before the changes are committed.

Files

The case is in the folder `pile_abutment` of `soil_FEM/paper/new_cases`, with the scripts `verify_element.py`, `run_case.py` and `compare.py` and the source PDF and digitised curves in `ref/`. The updated mesh check is `verify.py` in the folder `updated_mesh` next to it. The solver changes are in `fem/src/solve.rs` and `fem/src/beam.rs`, the phase switch in the app, the `updated_mesh` option of the MCP tools `add_phase` and `update_phase`, and `tools/build_linux.sh` builds the Linux binary that `cloud/vast_gpu.sh` uploads with `VAST_PREBUILT=1`.

References

Hetényi, M. (1946). *Beams on Elastic Foundation.* University of Michigan Press.

Hughes, T.J.R. and Winget, J. (1980). Finite rotation effects in numerical integration of rate constitutive equations arising in large-deformation analysis. *Int. J. Numer. Methods Eng.* 15(12), 1862–1867.

Mattiasson, K. (1981). Numerical results from large deflection beam and frame problems analysed by means of elliptic integrals. *Int. J. Numer. Methods Eng.* 17(1), 145–153.

PLAXIS (2012). *Embedded Pile Row in Plaxis 2D*, case study, November 2012.

Sluis, J. (2012). *Validation of Embedded Pile Row in Plaxis 2D*. MSc thesis, Delft University of Technology.

Sluis, J.J.M., Besseling, F. and Stuurwold, P.H.H. (2014). Modelling of a pile row in a 2D plane strain FE-analysis. *Numerical Methods in Geotechnical Engineering*, Hicks, Brinkgreve and Rohe (eds), Taylor and Francis, 277–282.